

Chapter 2 — Additional sampling techniques

Not every study has a complete frame or the budget for probability draws

Learning objectives

- Implement systematic sampling on an ordered frame with awareness of periodicity
- State when each design is defensible versus when generalizable claims are inappropriate

Beyond the three probability designs

- The previous part covered simple random, stratified
- Convenience and snowball sampling
- The decision rule is whether inference to a population is required or whether exploratory

Systematic sampling

- Systematic sampling selects every n th unit from an ordered frame after choosing a random
- If a population of five hundred must yield fifty interviews, the interval is ten
- Administration is often faster than drawing many independent random numbers

Example 2.13 — Systematic draw from a membership directory

- Example 2.13 — hands-on module
- Example 2.13 shows a clean use case and the periodicity trap on the same kind of list
- A professional society wants fifty interview slots from a five-hundred-person alphabetical
- With interval ten and a random start of four, members four, fourteen, twenty-four
- Shuffling before the draw removes that risk
- View files: ``modules/chapter2/example13/``

Convenience sampling

- Convenience sampling recruits whoever is easiest to reach
- Inclusion chances are unknown
- Convenience samples buy access at the cost of bias and limited diversity
- They scaffold instrument testing, not population claims

Example 2.14 — Convenience sample on a university campus

- Example 2.14 — hands-on module
- Example 2.14 shows a typical campus intercept study and why its estimates should be
- A researcher surveys the first one hundred students entering the library during exam week
- The design is cheap and fast, but evening commuters, remote students
- Explore the chapter example module
- View files: ``modules/chapter2/example14/``

Snowball sampling

- Snowball sampling starts with a few eligible seeds and asks them to refer others who meet
- The sample grows through social ties
- Referrals lower recruitment cost, yet they concentrate homogeneity
- Privacy protections for both participants and referrals are essential

Takeaways

- Use systematic sampling for efficient probability draws on clean lists
- Reserve convenience and snowball designs for exploratory or hard-to-reach settings
- When generalizable claims are required

Next

- Complete the quiz for this part
- Complete the quiz